

RITIKA BIST

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EDUCATION

M.Sc. Bioinformatics <i>Banasthali Vidyapith, Rajasthan</i>	2025 – 2027
B.Sc. Life Sciences <i>J.C. Bose University of Science and Technology, YMCA, Haryana</i>	2021 – 2024

RELEVANT COURSEWORK

In Progress / Planned: Machine Learning for Bioinformatics, Structural Bioinformatics

EXPERIENCE

Bioinformatics Research Trainee (Summer Training) — ICAR – Indian Agricultural Statistics Research Institute (ICAR-IASRI), New Delhi
4 May 2026 – 5 Jul 2026

Division of Agricultural Bioinformatics, under Dr. Anshuman Singh, Senior Scientist | Project: “Molecular Dynamics Simulation Study of PHT1 Phosphate Transporters: Structural Stability Analysis of Wild-type and SNP Mutant Variants”

- Ran all-atom molecular dynamics simulations (10 ns each, GROMACS) on 33 PHT1 phosphate transporter protein systems across three crop species — *Lupinus albus*, *Oryza sativa*, *Brassica napus* — comparing wild-type and SNP mutant variants using CHARMM36/TIP3P and GROMOS96 53a6/SPC force fields on the ASHOKA HPC cluster
- Designed and built a modular two-script Bash automation pipeline (11-stage GROMACS workflow plus a batch-execution wrapper) with stage-level file validation and fail-fast error handling, enabling unattended, reproducible MD runs across all 33 systems with zero stage failures
- Analyzed 66 backbone RMSD and radius-of-gyration trajectories to quantify structural stability differences between wild-type and mutant variants, identifying reproducible stabilizing and destabilizing SNPs (e.g., LaPHT1;2 F40L stabilizing; BnPHT1_3_3 Mutant A destabilizing) as candidates for experimental validation and marker-assisted breeding for phosphorus use efficiency
- Built working knowledge of GWAS methodology and statistical genetics through PhD-level lectures attended in a CSIR-funded research environment

PROJECTS

Transcriptomic Analysis of Therapy-Induced Senescence & PQQ Intervention — GSE290476 · github.com/sikkibist/transcriptomic-analysis-senescence

- Independently built a 7-phase bulk RNA-seq pipeline (FastQC → Trimmomatic → HISAT2 → StringTie → DESeq2) across 9 samples (85,321 genes), 91–92% mapping rate
- Identified 7,954 significant DEGs (bleomycin-induced senescence) and 2,579 DEGs (PQQ intervention); full visualization suite in R (PCA, volcano, MA, heatmap)

ATIP — AI-Assisted Transcriptomic Interpretation Platform — huggingface.co/spaces/sikkibist/ATIP · github.com/sikkibist/ATIP

- Built a Flask/Docker web tool that turns DESeq2 output into interactive volcano/MA/heatmap plots, Enrichr pathway enrichment, and PDF/DOCX reports
- Added an LLM interpretation layer (Groq API) to summarize enrichment results in plain language; deployed live on Hugging Face Spaces

Automated GROMACS Pipeline for PHT1 Molecular Dynamics — github.com/sikkibist/MD_pipeline

- Packaged the ICAR MD workflow into a standalone, reusable batch pipeline with per-stage validation and fail-fast error handling, validated across 33 protein systems

TECHNICAL SKILLS

Bioinformatics: NGS analysis, Differential expression, Molecular dynamics simulation, Pipeline development

Tools: FastQC, MultiQC, Trimmomatic, HISAT2, StringTie, DESeq2, GROMACS, BLAST, OMICSBOX

Programming: Python, R, Perl, Bash/Shell scripting

Systems: Linux/Unix, HPC environments, MySQL, XML, HTML